RoDo-EM: Robust Domain-Aware Entity Matching

Extended Technical Report*

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ABSTRACT

Entity Matching (EM) is a core task in data management that determines whether records from two or more data sources refer to the same real-world entity. Modern AI approaches, particularly those based on fine-tuning Pretrained Language Models (PLMs) and Large Language Models (LLMs), have achieved state-of-the-art performance on EM. Although these learned models achieve strong average performance, they often fall short on smaller or underrepresented domains (or categories). In this paper, we formalize the problem of robust domain-aware EM, where the objective is to improve underperforming domains while maintaining strong average-case performance. To this end, we introduce RoDo-EM, a robust optimization framework for domain-aware EM that combines robustness-aware loss functions, domain-aware sampling strategies, and regularization techniques to optimize a general-purpose EM model. We conduct extensive experiments on several domain-partitioned EM benchmarks and demonstrate that RoDo-EM consistently improves domain-level robustness while maintaining competitive overall matching performance, outperforming state-of-the-art EM methods across various robustness objectives.

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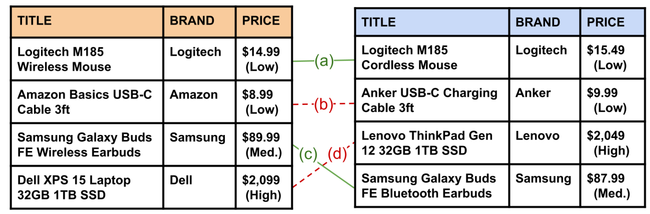

Figure 1: EM example with various price categories.

1 INTRODUCTION

Entity Matching (EM) is a longstanding problem in data integration and a fundamental operation in data processing pipelines. Also known as Entity Resolution or Record Linkage, EM is the task of determining whether a given pair of records refers to the same real-world object [12, 13]. These records may consist of structured rows in a table with a fixed schema or more loosely structured forms of data. Furthermore, the records may originate from the same table–for example, to perform deduplication–or from different sources, as in the traditional data integration setting [65].

EM has a rich history in the database community, with continual advances in methods, problem settings, and applications. Early EM research focused on traditional string-similarity and probabilistic approaches [18, 64], primarily for record linkage over relatively small databases. Over time, rule-based methods emerged [23, 33] followed by learning-based approaches [7, 8], and deep neural network models [17, 34]. More recently, the dominant paradigm has shifted towards fine-tuning pretrained models on task-specific data, particularly Pretrained Language Models (PLMs) [27], such as BERT [14] and RoBERTa [30], as well as Large Language Models (LLMs) [42], such as Llama [15]. Although LLMs have demonstrated strong zero-shot performance when labeled training data is scarce, fine-tuned PLMs remain the prevailing approach for EM [3], largely due to reduced inference latency and increased matching accuracy for targeted EM tasks.

In recent years, the increasing scale of pretrained models has enabled them to (1) achieve higher matching accuracy and (2) learn increasingly general representations of the EM task [11, 60]. This progress has led to the emergence of *domain-aware EM* [44, 52, 54], in which EM is viewed as a collection of related but distinct domains or subtasks. Existing work has explored domain adaptation, where a model learns to perform matching on a target domain by leveraging information from one or more source domains [52, 54]. More recently, researchers have also investigated task-specific EM [44], where multiple models are specialized for individual subtasks within a broader application domain (e.g., "Computers" and "Electronics" within the larger "E-commerce" domain). While these methods achieve state-of-the-art performance on domain-aware EM tasks, they rarely consider training a *single general-purpose EM* model that performs well across multiple domains [58]. In practice, such a model offers several advantages, including reduced maintenance, lower training cost, and decreased energy consumption compared to maintaining multiple specialized models.

LLMs in zero-shot settings can serve as general EM models across domains [11, 42]. However, these approaches present two important

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limitations. First, they incur substantially higher *inference latency, energy consumption, and deployment cost (e.g., when using hosted APIs)*, partially offsetting the practical advantages of a single general model. Second, and more importantly, they do not explicitly optimize for worst-case performance across domains [58, 59]. We use the term *robust* to refer to models that explicitly optimize the performance of their worst-performing domain, rather than only maximizing average matching performance. We note that this notion of robustness is inspired by a line of prior work in information retrieval that characterizes robustness in terms of a system's effectiveness on the most difficult topics [37, 57, 61]. Existing EM techniques, including the use of LLMs, may achieve strong overall performance while underperforming on smaller or underrepresented domains, providing no guarantees on worst-case domain performance.

In this paper, we study the problem of training a robust general-purpose domain-aware EM model that combines the efficiency of a PLM backbone with optimization objectives that account not only for overall matching performance, but also for performance on the most challenging domains. To motivate this problem setting, we first consider an illustrative example.

Example 1.1. *Consider a data engineer at a large e-commerce company who must match product listings against an existing product catalog. The workload consists of products from three price ranges: "Low" (<\$50), "Medium" (\$50–\$300), and "High" (>\$300), where low-priced products constitute the majority of the data and high-priced products represent a relatively small minority. For simplicity, we assume the listings and catalog share the same schema.*

The engineer first applies blocking *to eliminate unlikely matches, producing a set of* candidate pairs *for the matching stage. Figure 1 shows several representative candidate pairs: (a) and (b) belong to the "Low Price" domain, (c) to the "Medium Price" domain, and (d) to the "High Price" domain. A solid green line denotes a ground-truth match, while a dotted red line denotes a non-match.*

Having introduced domain-aware EM in the context of price ranges, we now motivate the need for *robust* optimization by illustrating why underperforming domains may be particularly important in practice.

Example 1.2. *The engineer's objective is not simply to train a model that achieves high matching accuracy* on average. *Such a model may perform exceptionally well on the abundant low-priced products while performing considerably worse on the relatively scarce high-priced products. In practice, however, the company is particularly concerned with ensuring that performance remains* robust *across all price ranges. Specifically, it seeks to maintain strong performance on the "High Price" domain, where incorrect predictions may have substantially greater financial consequences than errors on lower-priced products. For example, if pair (d) is incorrectly classified as a non-match, the company may lose the sale of a \$2,000 product. In contrast, making the same mistake on pair (a) may only affect a \$15 item. This example illustrates why maintaining strong performance on the "High Price" domain is important despite its relatively small representation in the training data.*

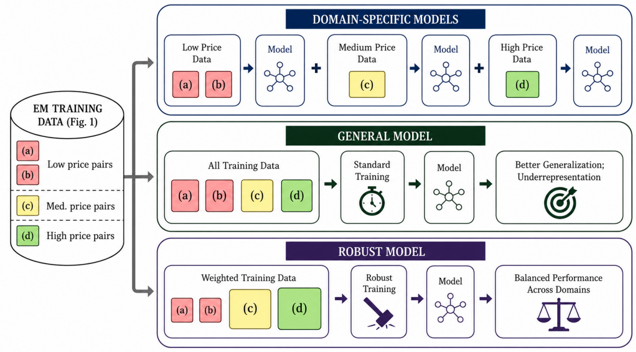

Figure 2: The specific, general, and robust EM models for the toy dataset of Figure 1.

There are several possible training strategies for the EM scenario introduced in Example 1.2, as illustrated in Figure 2. A straightforward approach is to train a separate PLM for each domain, fine-tuning one model for "Low Price," one for "Medium Price," and one for "High Price" (top of the figure). While intuitive, this approach requires training and maintaining multiple models, and domains with relatively few training samples, such as "High Price," may still exhibit poor matching performance. The middle portion of the figure illustrates an alternative strategy in which all labeled samples are used to train a single general-purpose model. Although this approach is simpler and allows information to be shared across domains, the resulting model may underperform on minority or inherently more challenging domains because optimization primarily favors overall performance. Finally, the bottom portion of the figure illustrates the proposed *robust* general model. By incorporating robust optimization techniques such as sample weighting and regularization, the model leverages all available training data while explicitly improving performance across domains despite differences in domain size and matching difficulty.

Consequently, the engineer in Example 1.2 would prefer to train this robustness-aware general model but must determine how best to optimize it to achieve strong performance across all domains. This is precisely the objective of our work: to develop a robust domain-aware EM framework, while also investigating alternative domain-level optimization objectives. In pursuit of this goal, our paper makes the following contributions:

- *Problem Formulation (Section 3).* We formalize the problem of *Robust Entity Matching Across Domains* (REMAD), extending traditional EM to account for general-purpose models that optimize domain-level robustness across multiple domains.
- *Robust Optimization for EM (Section 4).* We investigate several robust optimization techniques for domain-aware EM, including Distributionally Robust Optimization (DRO) [6], and study their effectiveness for improving performance.
- *RoDo-EM (Section 4.3).* We introduce **RoDo-EM**, a modular framework for robust domain-aware entity matching that combines robustness-aware optimization objectives, domain-aware sampling strategies, and regularization techniques. Its configurable design enables practitioners to tailor

the optimization pipeline to the robustness requirements of their application.

- *Experimental Evaluation (Sections 5-6).* We evaluate robust optimization by fine-tuning PLMs on general EM tasks and comparing against state-of-the-art baselines. To support evaluation, we derive 4 general EM benchmark tasks with diverse domain distributions, which we also release to facilitate future research on robust domain-aware EM.

2 RELATED WORK

Our work builds upon recent advances in machine learning and deep learning for Entity Matching (EM) [4, 17, 34], particularly approaches based on Pretrained Language Models (PLMs) [3, 27] and Large Language Models (LLMs) [42, 51]. While these learned techniques have also been applied to other stages of the Entity Resolution pipeline, such as blocking [9, 24], this work focuses exclusively on the matching stage.

Among PLM-based approaches, DITTO [27] was one of the first methods to demonstrate the effectiveness of fine-tuning pretrained language models for EM. Since its introduction, the DITTO pipeline has served as the foundation for subsequent systems [10, 20, 21, 40, 68], including AdapterEM [35], MultiMatch [36] and BEACON [44]. Accordingly, we adopt the DITTO implementation as the foundation for our experimental framework.

More recently, Large Language Models (LLMs) have emerged as a promising approach for EM. Early work explored both open-source and hosted LLMs for zero-shot and few-shot matching [42], while subsequent studies proposed fine-tuned LLMs for improved task-specific performance [51, 67]. Prompting-based methods have likewise become increasingly popular, leveraging in-context learning, retrieval augmentation, and specialized prompting strategies to improve matching accuracy [11, 19, 22, 62]. Although these approaches have demonstrated improved matching accuracy in several settings, their higher inference latency and computational cost have thus far limited widespread adoption in practice [3].

Domain-Aware Entity Matching. Recent work has explored domain-aware learning for EM, where a domain corresponds to a particular type of data on which the model performs matching [56, 66]. The predominant line of research in this area is *domain adaptation*, where a model learns to perform EM on a target domain with limited or no labeled data by leveraging a source domain with abundant labeled examples. Early work focused on deep learning approaches, including Thirumuruganathan et. al. [53], who proposed a distributed representation for record pairs, and Kasai et. al. [25], who combined transfer learning with active learning to address low-resource EM settings. DAME [54] introduced a mixture-of-experts framework to capture domain-specific knowledge from multiple source domains and transfer it to a target domain. DADER [55] further formalized domain adaptation for EM as an end-to-end framework by introducing a feature alignment module designed to reduce distributional shift between source and target domains. More recently, MFSN [52] distinguishes between shared and domain-specific matching features, while Okayama et. al. [39] investigate domain adaptation for prompt-based and LLM-based EM.

Although these methods demonstrate the importance of domain structure in EM, the domain adaptation setting is fundamentally different from the *EM across domains* problem considered in this paper. Domain adaptation focuses on transferring knowledge from one or more source domains (e.g., "Computers") to a target domain with limited labeled data (e.g., "Cell Phones"). In contrast, EM across domains seeks to leverage information from all available data to improve matching performance across the entire collection of domains. A recent work in this setting, BEACON [44], formally introduced the EM across domains problem and proposed distribution-aware sampling strategies for training domain-specific EM models under a fixed labeling budget. Our work builds upon the conceptual foundation established by BEACON, but addresses a fundamentally different optimization problem. Rather than selecting training data for multiple domain-specific models, we investigate how to train a single general-purpose EM model whose performance remains robust across all domains. In particular, we study optimization objectives that improve lower-bound domain performance and domain parity while preserving strong overall matching performance.

Robust Optimization Techniques. Robust optimization for EM has received relatively limited attention [1, 27, 32]. Existing approaches primarily define robustness as strong performance in the context of label imbalance and distribution shift in EM datasets. To the best of our knowledge, however, no prior work has explicitly investigated domain or group-level robustness for EM, which is the focus of this paper. In contrast, robust optimization across groups has become an active area of research in the broader machine learning community [16, 28, 38, 47].

Sagawa *et al.* [47] introduced Group Distributionally Robust Optimization (Group DRO), extending classical Distributionally Robust Optimization (DRO) [6] to optimize worst-group performance during training. They further demonstrated that regularization and early stopping are critical for achieving strong robustness across groups. Liu et. al. [28] extended this framework to settings where group labels are unavailable by learning to identify minority groups during training. More recently, Nguyen et. al. [38] studied Group DRO in the context of domain generalization, introducing Risk Distribution Matching as a mechanism for encouraging balanced performance across domains.

Our work builds upon this line of research by applying group-based robust optimization techniques to the EM across domains setting. Specifically, we investigate how robust optimization can be used to train a single general-purpose EM model that improves worst-domain performance while maintaining strong overall matching accuracy.

Algorithmic Fairness. Closely related to robust group optimization is the problem of algorithmic fairness. Fairness has been studied extensively in the machine learning literature, particularly in recent years [5, 31, 46, 63]. As AI systems are increasingly deployed in real-world decision-making, ensuring that models treat different demographic groups equitably has become an important research objective. Fairness methods commonly seek some notion of group parity, where groups are defined by sensitive attributes such as race, gender, or age.

Because EM often involves records describing people and may influence downstream decision-making, fairness has also received growing attention within the EM community. Araujo *et al.* [2] provide a comprehensive survey of fairness and explainability throughout the Entity Resolution pipeline. Shahbazi *et al.* [50] explicitly evaluate EM systems using fairness-oriented metrics such as precision and recall parity across sensitive groups. More recently, FairNM [29] considered the related task of *Name Matching*, where the objective is to reconcile variations in personal names arising from formatting differences, typographical errors, or misspellings while simultaneously optimizing a fairness objective.

Our work is related to fairness in both machine learning and EM but differs in two important respects. First, fairness research typically focuses on groups defined by sensitive attributes. In contrast, this work considers non-sensitive application domains, such as product categories or price ranges, where the objective is to improve performance across operational domains rather than demographic groups. Second, although RoDo-EM includes domain-parity objectives inspired by the fairness literature, parity is only one notion of robustness within our framework. Depending on the application, practitioners may want to optimize for worst-case domain performance or domain parity while maintaining strong overall matching accuracy.

3 PRELIMINARIES

We now introduce the notation and formal definitions used throughout the remainder of the paper (Section 3.1), formulate the standard EM task (Section 3.2), extend it to the domain-aware EM setting (Section 3.3), and formally define the problem of Robust Entity Matching Across Domains (REMAD, Section 3.4).

3.1 Notations and Blocking

Following Li *et al.* and subsequent work [27, 44], we define EM as the task of matching records between two distinct datasets. Alternative formulations also exist, such as matching records within a single dataset for deduplication [49]. Let D_{left} and D_{right} denote the two datasets to be matched. We assume that each dataset is tabular and consists of *entities* (or records), where an entity r is represented as a row of structured information. We focus on *clean-clean* setting, which assumes that neither D_{left} nor D_{right} contain duplicate records internally. The set of all possible record pairs is given by the Cartesian product $D_{\text{left}} \times D_{\text{right}}$. We refer to these pairs as *candidates*, and to the complete collection of pairs as the *candidate set.*

The EM pipeline typically consists of two stages: blocking followed by matching. During the blocking phase, lightweight heuristics are applied to eliminate unlikely matches from the candidate set, thereby reducing the search space. This process produces a subset of candidates $N \subset D_{\text{left}} \times D_{\text{right}}$ such that $|N| \approx \max(O(|D_{\text{left}}|), O(|D_{\text{right}}|))$, avoiding the $O(|D_{\text{left}}| \times |D_{\text{right}}|)$ complexity of exhaustive pairwise comparison. While blocking is essential for scalability (and may also be used to construct domain partitions), the focus of this paper is on improving the downstream matching stage.

We assume that the candidate set N can be partitioned into j disjoint subsets, each corresponding to a distinct domain, where

$$N = n_1 \cup n_2 \cup \cdots \cup n_j, \quad n_i \cap n_k = \emptyset \text{ for } i \neq k.$$

We assume that each candidate pair $(r_1, r_2) \in n_i$ consists of two records that both belong to domain i. Thus, domains partition candidate pairs rather than individual records. Throughout this work, we consider product data, where domains may correspond to product categories (e.g., “Computers,” “Clothing”), brands (e.g., “Apple,” Samsung”), or price ranges (e.g., “High,” “Low”).

3.2 Entity Matching

Given a filtered candidate set N, matching determines which candidate pairs correspond to matching entities. This is formulated as a binary classification problem: for each candidate pair $(r_1, r_2)_i \in N$, we assign a label $y_i \in 0, 1$, where $y_i = 1$ if r_1 and r_2 refer to the same real-world entity, and $y_i = 0$ otherwise. We refer to a candidate pair together with its corresponding label as a *sample.*

To perform this classification, we adopt PLMs such as BERT [14], RoBERTa [30], and DistilBERT [48]. A PLM is a transformer-based model pretrained on a large text corpus and subsequently fine-tuned for downstream tasks such as EM. Following the DITTO framework [27], an entity r with p attributes is serialized as:

$$\texttt{[COL] attr}_1 \texttt{ [VAL] val}_1 \cdots \texttt{ [COL] attr}_p \texttt{ [VAL] val}_p$$

and each candidate as:

$$\texttt{[CLS]}\ r_1\ \texttt{[SEP]}\ r_2$$

The serialized candidate pair is then provided as input to the PLM. The `[SEP]` token marks the boundary between the left and right entities, while the `[CLS]` token serves as a learned representation of the entire candidate pair. The resulting `[CLS]` embedding is passed through a fully connected layer, followed by a sigmoid or softmax activation function, to produce the predicted match label $\hat{y}_i$.

3.3 Entity Matching Across Domains

While traditional EM optimizes performance over the entire candidate set, this work explicitly considers the partitioning of candidates into domains and studies optimization objectives defined over these domain-level partitions. Following BEACON [44], we denote the set of domains by $\mathcal{D}$, where $|\mathcal{D}| = j$. Given this formulation, there are two primary approaches for incorporating domain information during training: (1) training separate *domain-specific* EM models for each domain or (2) training a single *general-purpose* EM model using data from all domains.

For the domain-specific approach, we train j independent models, where the training set X_i for domain $i \in \mathcal{D}$ is given by

$$X_i = n_i.$$

That is, each model is trained exclusively on the samples belonging to its corresponding domain. As discussed in Section 1, this approach requires maintaining one model per domain and may underperform when certain domains contain relatively few labeled examples.

In this paper, we instead focus on training a *single general-purpose* EM model. In this setting, the single training set X consists of the union of all domain partitions:

$$X = N = n_1 \cup n_2 \cup \cdots \cup n_j.$$

Consequently, a single model is trained using samples from all the domains. While this approach improves scalability and enables information sharing across domains, it also introduces a new challenge. Different domains often exhibit distinct schema structures, attribute distributions, data quality, and matching characteristics, causing some domains to be inherently more difficult than others. For example, certain domains may contain noisier attribute values, greater lexical ambiguity, or more subtle distinctions between matching and non-matching records. As a result, a general-purpose model may achieve strong overall performance while consistently underperform on these more challenging domains, even when they are well represented in the training data. Addressing this challenge is the primary motivation for the robust optimization framework.

3.4 Problem Definition

We now extend the formulation of training a general-purpose domain-aware EM model to the *Robust Entity Matching Across Domain* (REMAD) setting. Let θ denote the parameters of the general-purpose EM model, and X the set of labeled pairwise samples available for training. Commonly, a cross-entropy loss is used to perform binary classification over X to optimize θ, and we denote this loss as $\mathcal{L}(\theta)$. The corresponding F1 score, the standard evaluation metric used in EM, is denoted by $F(\theta)$. We further define the corresponding domain-level metrics. For each domain $i \in \mathcal{D}$, we define $\mathcal{L}_i(\theta)$ as the average binary cross-entropy loss over the samples in n_i, and $F_i(\theta)$ as the corresponding domain-level F1 score.

Definition 1 (Robust Entity Matching Across Domains (REMAD))**.** Given a training set X consisting of a domain partition based on $\mathcal{D}$, the objective of REMAD is to train a robust general-purpose EM model θ^* such that

$$\theta^* = \arg\max_{\theta} \min_{i \in \mathcal{D}} F_i(\theta); \quad F(\theta^*) \approx \max_{\theta} F(\theta).$$

Our goal is therefore to *optimize the worst-domain performance of a general-purpose EM model while preserving strong overall matching performance.* This formulation seeks a model that is both more robust across domains and more reliable in practical deployment.

In practice, the solution θ^* is obtained by optimizing differentiable loss functions defined over the domain-level losses $\mathcal{L}_i(\theta), \forall i \in \mathcal{D}$. Section 4 presents several such optimization strategies, including worst-k optimization, Group DRO, and uniform weighting.

4 ROBUST DOMAIN-AWARE ENTITY MATCHING (RODO-EM)

We now present the optimization framework used to enforce robustness in domain-aware EM. We begin by generalizing robustness to a family of domain-level optimization objectives and introduce corresponding loss functions (Section 4.1). Next, Section 4.2 presents domain-aware sampling strategies that further encourage robust performance during training. Section 4.3 then introduces RoDo-EM (**Ro**bust **Do**main-Aware **E**ntity **M**atching optimizer), a unified framework for optimizing robust EM performance across domains.

4.1 Domain-Level Optimization Objectives

Although REMAD formally defines robustness as maximizing the performance of the worst-performing domain, the most appropriate domain-level objective ultimately depends on the requirements of a particular application. Accordingly, we generalize robustness to two major categories of domain-level optimization objectives: *performance-based* objectives and *parity-based* objectives. Performance-based objectives emphasize the predictive performance of a particular subset of domains, whereas parity-based objectives seek to reduce disparities in performance across all domains. Although these objective families are implemented differently, both seek to produce a robust EM model under a particular domain-level criterion.

For each optimization objective, we first define a target evaluation metric $m : \Theta \to \mathbb{R}$ that specifies the desired domain-level property (e.g., worst-domain F1 or F1 entropy). We then define a corresponding differentiable training loss $\mathcal{L} : \Theta \to \mathbb{R}$ that serves as a surrogate objective for optimizing this metric during gradient-based training. Because many evaluation metrics used throughout this paper (e.g., F1, precision, recall, and the associated parity metrics) depend on thresholded predictions and are therefore non-differentiable, the corresponding loss functions optimize differentiable approximations rather than the metrics themselves. For each domain $i \in \mathcal{D}$, we assume access to the domain-level loss $\mathcal{L}_i(\theta)$ and differentiable soft counterparts to F1, precision, and recall, denoted by $\tilde{F}_i(\theta)$, $\tilde{P}_i(\theta)$, and $\tilde{R}_i(\theta)$, respectively. These surrogate metrics are computed by replacing thresholded predictions with the model's predicted positive-class probabilities, yielding differentiable approximations that are used exclusively during optimization. For example, we estimate the soft true positives for domain i as

$$\tilde{TP}_i = \sum_{(x,y) \in \tilde{n}} p(x)y$$

where $p(x)$ denotes the model's predicted probability that sample x in the current minibatch $\tilde{n}$ belongs to the positive class. Soft precision, recall, and F1 are then defined analogously by replacing TP, FP, and FN with their soft counterparts. Once training is complete, these surrogate metrics are discarded and the original thresholded metrics are used for evaluation.

Performance-based Objectives. The most common objective in robust domain-aware optimization is to improve the k worst-performing domains, i.e., the domains with the lowest F1 scores. This objective encourages the model to allocate additional optimization effort toward the domains it currently finds most difficult, thereby improving worst-case robustness. In Example 1.1, we motivated the Worst-1 objective using a toy e-commerce dataset in which a minority domain of high-priced products contains more challenging matching relationships and incurs greater business costs when matching errors occur. We define the worst-k metric as the average F1 score over these domains:

$$m_{\text{worst-}k}(\theta) = \frac{1}{k} \sum_{i \in \text{BottomK}_{i \in \mathcal{D}}(F_i(\theta))} F_i(\theta). \quad (1)$$

The objective is to maximize $m_{\text{worst-}k}(\theta)$. This formulation can also extend to the best and median-performing domains by replacing BottomK with the corresponding domain-selection operator.

To optimize the Worst-k objective, we define a weighted loss that places additional emphasis on the selected domains:

$$\begin{aligned} \mathcal{S}_{\text{worst-}k} &= \text{BottomK}_{i \in \mathcal{D}}(\mathcal{L}_i(\theta)), \\ \mathcal{L}_{\text{worst-}k}(\theta) &= \sum_{i \in \mathcal{S}_{\text{worst-}k}} \gamma_{\text{worst-}k} \mathcal{L}_i(\theta) + \sum_{i \notin \mathcal{S}_{\text{worst-}k}} \mathcal{L}_i(\theta), \end{aligned} \quad (2)$$

where $\gamma_{\text{worst-}k} > 1$ controls the additional emphasis assigned to the objective domains relative to the remaining domains. Consequently, $\mathcal{L}_{\text{worst-}k}$ increases the optimization effort devoted to the current domains with the highest loss while still allowing all other domains to contribute to the training objective.

We additionally consider a loss inspired by Group Distributionally Robust Optimization (Group DRO) [6, 47]. Distributionally Robust Optimization minimizes the worst-case expected loss over an uncertainty set of data distributions, while Group DRO instantiates these distributions as predefined groups, which naturally correspond to domains in our setting.

Unlike the formulation of Sagawa *et al.* [47], which explicitly maintains adversarial group weights that increasingly concentrate on the highest-loss groups throughout training. We employ a smooth log-sum-exp objective that emphasizes higher-loss domains while retaining contributions from all domains. The resulting loss is

$$\mathcal{L}_{\mathrm{DRO}}(\theta) = \frac{1}{\gamma_{\mathsf{DRO}}} \log \left(\sum_{i \in \mathcal{D}} \exp(\gamma_{\mathsf{DRO}} \mathcal{L}_i(\theta)) \right), \tag{3}$$

where $\gamma_{\mathsf{DRO}} > 0$ controls the degree of robustness. As γ_{DRO} increases, the objective places progressively greater emphasis on higher-loss domains. Smaller values approach a uniformly weighted objective.

Parity-based Objectives. Parity-based objectives seek to reduce disparities across domains rather than emphasizing a particular subset of domains. Because factors such as domain difficulty and training-set size can naturally produce uneven performance, a practitioner may prefer an objective that treats every domain equally.

We define a *uniform* objective that maximizes the macro-average domain F1 score:

$$m_{\mathrm{uniform}}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} F_i(\theta). \tag{4}$$

The corresponding training loss assigns equal weights:

$$\mathcal{L}_{\mathrm{uniform}}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \mathcal{L}_i(\theta). \tag{5}$$

Unlike the standard sample-level cross-entropy objective, which implicitly gives greater influence to domains containing more training samples, the uniform loss gives every domain equal importance. Consequently, the training objective becomes a macro-average over domains rather than a micro-average over individual samples.

Maximizing average domain performance alone, however, does not guarantee that performance is evenly distributed. A model may achieve a high macro-average F1 score while still containing one or more domains whose performance differs substantially from the others. We therefore define four additional parity metrics that can be optimized jointly with average predictive performance: normalized F1 entropy, F1 variance, Positive Predictive Value Parity (PPVP), and True Positive Rate Parity (TPRP). PPVP and TPRP correspond to parity in domain-level precision and recall, respectively, following prior work on fairness in EM [50].

For F1 entropy, we first normalize the domain-level F1 scores:

$$q_i(\theta) = \frac{F_i(\theta)}{\sum_{z \in \mathcal{D}} F_z(\theta)}. \tag{6}$$

We then define the normalized F1 entropy as

$$m_{\mathrm{entropy}}(\theta) = -\frac{1}{\log |\mathcal{D}|} \sum_{i \in \mathcal{D}} q_i(\theta) \log q_i(\theta). \tag{7}$$

The normalization constrains $m_{\mathrm{entropy}}(\theta)$ to $[0, 1]$, with larger values indicating that F1 performance is more evenly distributed across domains.

Let

$$\overline{F}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} F_i(\theta), \quad \overline{P}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} P_i(\theta),$$

and

$$\overline{R}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} R_i(\theta)$$

denote the corresponding domain-level means. We define the remaining parity metrics as the negative variance of the relevant performance measure:

$$m_{\mathrm{var}}(\theta) = -\frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \left(F_i(\theta) - \overline{F}(\theta) \right)^2, \tag{8}$$

$$m_{\mathrm{PPVP}}(\theta) = -\frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \left(P_i(\theta) - \overline{P}(\theta) \right)^2, \tag{9}$$

$$m_{\mathrm{TPRP}}(\theta) = -\frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \left(R_i(\theta) - \overline{R}(\theta) \right)^2. \tag{10}$$

Because these metrics are defined as negative variances, values closer to zero indicate greater parity. Defining all four parity measures as higher-is-better metrics allows them to be incorporated consistently into the training objective. However, in Section 6 we report the positive counterparts for consistency.

For each parity metric, we define a differentiable surrogate by replacing the thresholded domain-level F1, precision, and recall values with their corresponding soft counterparts. Consequently, the surrogate metrics have the same functional form as Equations (7)–(10), differing only in that F_i, P_i, and R_i are replaced by $\tilde{F}_i$, $\tilde{P}_i$, and $\tilde{R}_i$, respectively. For example, let

$$\overline{\tilde{F}}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \tilde{F}_i(\theta).$$

The differentiable surrogate for F1 variance is then defined as

$$\tilde{m}_{\mathrm{var}}(\theta) = -\frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \left(\tilde{F}_i(\theta) - \overline{\tilde{F}}(\theta) \right)^2. \tag{11}$$

We then optimize the following parity-regularized loss for a given metric “par”:

$$\mathcal{L}_{\mathrm{par}}\left(\theta; \tilde{m}_{\mathrm{par}}\right) = \mathcal{L}(\theta) - \gamma_{\mathrm{par}} \tilde{m}_{\mathrm{par}}(\theta), \tag{12}$$

where $\gamma_{\mathrm{par}} > 0$ controls the trade-off between overall predictive performance and domain parity. Larger values place greater emphasis on improving domain parity, whereas smaller values prioritize minimizing the average prediction loss. By jointly optimizing prediction loss and parity, this formulation encourages more uniform performance across domains while maintaining overall accuracy.

4.2 Domain-Aware Batch Sampling

As an additional component of the REMAD optimization framework, we consider how the composition of each training batch can be used to further emphasize the training objectives introduced in Section 4.1. While the loss functions we defined determine how strongly each domain contributes to the optimization objective, the sampling strategy determines how frequently samples from each domain are presented to the model during training.

Consider a mini-batch constructed at training iteration e. We define the batch sampling distribution by a probability distribution P_{opt} over the domains, where $P_{\text{opt}}(i)$ denotes the probability that a sample selected for the mini-batch is drawn with replacement from domain i under sampling strategy *opt*. Different sampling strategies therefore correspond to different choices of P_{opt}, allowing the training process to emphasize different subsets of domains.

Balanced Batches. A simple strategy is to enforce equal representation from every domain within each mini-batch. To achieve this, domains are sampled with uniform probability:

$$P_{\text{BAL},e}(i) = \frac{1}{|\mathcal{D}|}.$$

Consequently, each domain contributes approximately the same number of samples to every mini-batch. This strategy prevents large domains from dominating individual optimization steps and naturally complements the uniform weighting objective.

Sampling by Loss. Rather than sampling every domain equally, the sampling distribution can adapt dynamically according to the current performance of each domain. Let $\widehat{\mathcal{L}}_i(\theta_e)$ and $F_i(\theta_e)$ denote the exponentially smoothed domain-level loss and domain-level F1 score, respectively, after epoch e.

When sampling according to loss, each training sample from domain i is assigned a weight $\widehat{\mathcal{L}}_i(\theta_e)$. Consequently, the probability of selecting samples from domain i is

$$P_{\text{Loss},e}(i) = \frac{|n_i|\widehat{\mathcal{L}}_i(\theta_e)}{\sum_{k\in\mathcal{D}}|n_k|\widehat{\mathcal{L}}_k(\theta_e)},$$

where $|n_i|$ denotes the number of training samples in domain i. Thus, domains with larger losses receive proportionally more samples during subsequent training iterations, although their total sampling probability also depends on domain size.

Similarly, sampling can be based on domain-level F1 scores by assigning a weight of $1-F_i(\theta_e)$ to each training sample from domain i, giving greater probability to domains with lower performance:

$$P_{\text{F1},e}(i) = \frac{|n_i|\,(1-F_i(\theta_e))}{\sum_{k\in\mathcal{D}}|n_k|\,(1-F_k(\theta_e))}.$$

Unlike balanced sampling, these adaptive sampling strategies evolve throughout training, allowing the model to allocate additional optimization effort toward domains that are currently performing poorly.

4.3 RoDo-EM

Having introduced the optimization objectives and domain-aware sampling strategies used throughout this work, we now present RoDo-EM (**Ro**bust **Do**main-Aware **E**ntity **M**atching optimizer), our unified framework for robust optimization in domain-aware EM.

Definition 2. RoDo-EM is a configurable optimization framework that combines a backbone EM model with modular robustness components to produce a robust domain-aware EM training pipeline. A RoDo-EM configuration is formally defined as

$$\texttt{RoDo-EM} = (\mathcal{O}, \mathcal{L}, \mathcal{P}, \mathcal{R}, \mathcal{M}),$$

where $\mathcal{O}$ denotes the optimization objective, $\mathcal{L}$ the robust loss function, $\mathcal{P}$ the domain-aware sampling strategy, $\mathcal{R}$ the set of chosen regularization strategies, and $\mathcal{M}$ the backbone EM model.

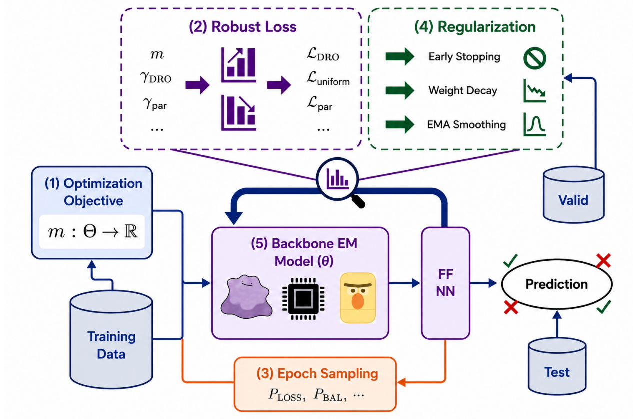

Figure 3: Overview of the RoDo-EM framework.

Figure 3 illustrates the RoDo-EM framework. Training begins by specifying (1) a domain-level optimization objective, which is controlled by a metric m according to the desired notion of robustness (e.g., worst-performing, F1 variance, PPVP variance, etc.). This objective defines (2) a corresponding robust loss function, such as the weighted Top-k, Group DRO, or uniform objective introduced in Section 4.1. During optimization, (3) a domain-aware sampling strategy from Section 4.2 determines how training samples are selected for each mini-batch according to the chosen objective (e.g., balanced sampling, loss-based sampling, or size-based sampling).

These optimization components operate jointly with (4) standard regularization techniques and (5) a backbone EM model. Specifically, we incorporate early stopping based on validation performance, L2 weight decay to discourage overly complex models, and Exponential Moving Average (EMA) smoothing to stabilize optimization. Together, these regularization techniques improve generalization and complement the robust optimization objectives. The resulting training procedure can be applied to any supervised EM backbone, such as a PLM or a fine-tuned LLM. After training, the best-performing model is selected based on validation set performance and evaluated on the domain-partitioned test data.

A key characteristic of RoDo-EM is its modularity. In practice, applications differ in their robustness requirements, computational constraints, and available backbone models. RoDo-EM is thus designed to allow each component of the optimization pipeline to be selected independently. Different optimization objectives, robust loss functions, sampling strategies, regularization methods, and backbone models can be combined to accommodate diverse application requirements without modifying the overall framework.

Accordingly, in our experimental evaluation, the hyperparameters associated with each optimization configuration (e.g., γ, $\gamma_{\texttt{DRO}}$, $\gamma_{\texttt{par}}$, learning rate, weight decay, and sampling strategy) are selected independently using grid search on the validation set.

5 EXPERIMENTAL SETUP

We now turn to a comprehensive experimental evaluation to assess the effectiveness of the proposed robust domain-aware optimization techniques. Section 5.1 introduces the benchmark datasets and domain partitions used throughout the evaluation. Section 5.2 describes the implementation details, and Section 5.3 presents the baseline methods against which we compare our approach.

5.1 Datasets

To evaluate the REMAD setting, we construct evaluation tasks from four widely used EM benchmarks: the WDC Multi-Dimensional Entity Matching Benchmark (WDC-MD) [41], the WDC Products Corpus [43], Abt-Buy [26], and Walmart-Amazon [34].

Domain Partitioning. We partition each benchmark into multiple domains according to a semantic attribute. In practice, this can be viewed as grouping candidate pairs by a blocking key. When necessary, we retain only domains containing at least 100 samples to ensure sufficient data for domain-aware training and evaluation. Collectively, these datasets exhibit diverse domain distributions, allowing us to evaluate REMAD under a variety of scenarios.

The WDC Multi-Dimensional Benchmark (WDC-MD). WDC-MD [41] is a large product matching benchmark containing datasets that vary along three dimensions: the percentage of corner cases, the percentage of unseen entities in the test set, and the dataset size. Throughout this paper, we use the large variant with the 50% corner-case and 50% unseen-entity setting, providing sufficient samples for each domain while representing the benchmark's central evaluation configuration.

We partition WDC-MD using its provided *product category* labels. This produces a right-skewed distribution across 12 domains, with several minority categories (e.g., "Automotive" and "Musical Instruments") and a few dominant categories (e.g., "Computers" and "Office Products"). Consequently, WDC-MD provides a challenging REMAD setting in which minority domains are particularly susceptible to being overshadowed during training, making it well suited for evaluating robust optimization techniques.

The WDC Products Corpus. We also include the medium variant of the WDC Products Corpus [43], which represents a more balanced distribution than the above WDC-MD. Similar to WDC-MD, this benchmark consists of product matching data partitioned by product category. The corpus contains four categories–"Cameras," "Computers," "Shoes," and "Watches"–whose domain sizes range from approximately 4k to 6k samples. Compared to WDC, these domains are considerably more balanced, allowing us to evaluate REMAD when all domains are well represented in the training data.

The Abt-Buy Benchmark. The Abt-Buy benchmark [26] is a widely used EM benchmark that aligns product listings from abt.com with listings from buy.com. Compared to the WDC datasets, Abt-Buy contains a more text-heavy and low-attribute schema, allowing us to evaluate REMAD in settings where domain boundaries arise from heterogeneous textual representations rather than standardized attributes. We partition Abt-Buy by *brand*, producing 10 domains with a similarly right-skewed distribution.

The Walmart-Amazon Benchmark. Finally, we use the Walmart-Amazon benchmark [34] for the case study presented in Section 6.4. Walmart-Amazon contains product listings collected from the Walmart and Amazon e-commerce platforms. As motivated in Examples 1.1 and 1.2, we use price as the domain attribute to demonstrate how REMAD can improve practical EM deployments.

Table 1: Evaluation Metrics for REMAD.

Metric	Equation	Optimized by
Weighted F1	$\bar{F}_{\text{weighted}}(\theta) = \frac{\sum_{i \in \mathcal{D}} F_i(\theta)\,\lvert t_i \rvert}{\sum_{i \in \mathcal{D}} \lvert t_i \rvert}$	Baseline
Macro F1	$\bar{F}_{\text{macro}}(\theta) = \frac{1}{\lvert\mathcal{D}\rvert} \sum_{i \in \mathcal{D}} F_i(\theta)$	$\mathcal{L}_{\text{uniform}}$
Minimum-k F1	$\frac{1}{k} \sum_{i \in \text{BottomK}_{i \in \mathcal{D}}(F_i(\theta))} F_i(\theta)$	$\mathcal{L}_{\text{worst-}k}$, $\mathcal{L}_{\text{DRO}}$
F1 Entropy	$-\frac{1}{\log \lvert\mathcal{D}\rvert} \sum_{i \in \mathcal{D}} q_i(\theta) \log q_i(\theta)$	$\mathcal{L}_{\text{par}}(\theta; m_{\text{entropy}})$
F1 Variance	$\frac{1}{\lvert\mathcal{D}\rvert} \sum_{i \in \mathcal{D}} \left(F_i(\theta) - \bar{F}_{\text{macro}}(\theta)\right)^2$	$\mathcal{L}_{\text{par}}(\theta; m_{\text{var}})$
PPVP Variance	$\frac{1}{\lvert\mathcal{D}\rvert} \sum_{i \in \mathcal{D}} \left(P_i(\theta) - \bar{P}(\theta)\right)^2$	$\mathcal{L}_{\text{par}}(\theta; m_{\text{PPVP}})$
TPRP Variance	$\frac{1}{\lvert\mathcal{D}\rvert} \sum_{i \in \mathcal{D}} \left(R_i(\theta) - \bar{R}(\theta)\right)^2$	$\mathcal{L}_{\text{par}}(\theta; m_{\text{TPRP}})$

Specifically, we partition the dataset into three price-based domains ("Low," "Medium," and "High") using representative price thresholds. Because the two products in a candidate pair may differ in price, each pair is assigned to a domain according to the average price of its two products. As a result, the domain distribution mirrors the motivating example, with low-priced products forming the largest domain, followed by medium and high-priced products.

5.2 Implementation Details

All experiments were conducted on NVIDIA A100 (80GB) GPUs within a high-performance computing cluster running Linux and Python 3.7.[2] Unless otherwise stated, all experiments fine-tune DistilBERT [48] as the PLM backbone. We adopt the dataset-specific hyperparameter settings from DITTO [27]. Robustness-specific hyperparameters ($\gamma, \gamma_{\texttt{DRO}}, \gamma_{\texttt{Size}}, \eta, \omega$) are selected independently for each optimization method using a grid search.

We next describe the evaluation metrics used throughout our experimental study. Each optimization objective introduced in Section 4.1 has a corresponding evaluation metric. Let $F_i(\theta)$ denote the F1 score achieved on domain $i \in \mathcal{D}$, and let t_i denote the number of test samples in that domain. Table 1 summarizes the evaluation metrics reported throughout the remainder of the paper.

In the standard EM setting, models are typically evaluated using **weighted F1**, which computes a weighted average of the domain-level F1 scores according to the number of samples in each domain. We additionally report **macro F1**, which assigns equal weight to every domain regardless of its size. Table 1 also formalizes the Minimum-k F1 scores, and other parity metrics introduced in Section 4 such as F1 Entropy, F1 Variance, PPVP, and TPRP.

5.3 Baselines

To evaluate the effectiveness of RoDo-EM, we compare against several recent EM methods spanning general-purpose, domain-specific, graph-based, budget-aware, and prompt-based approaches. Unless otherwise stated, all PLM-based methods use the same backbone

[2] See https://github.com/nbpulsone/RODO-EM for open-source code and data.

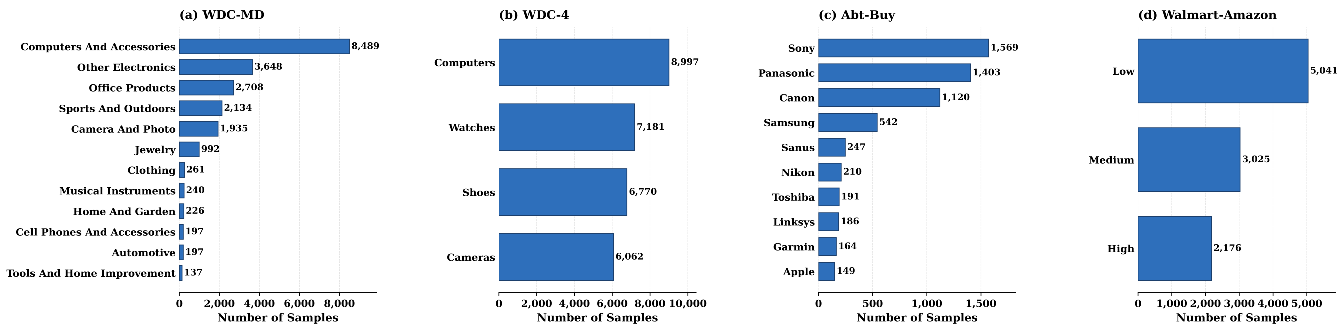

Figure 4: The distribution of samples across domains for each domain-partitioned benchmark.

Table 2: Robustness metrics across baselines. RoDo-EM is compared with other modern EM methods over three benchmark datasets.

(a) WDC-MD (50% cc, 50% unseen)

Model	Min. ↑	Macro ↑	Weighted ↑	F1 Entropy ↑	PPVP Var. ↓
GEN (Ditto)	0.421	0.815	**0.810**	0.993	0.026
SPEC (Ditto)	0.240	0.506	0.616	0.978	0.052
FlexER	0.491	0.656	0.658	0.993	0.024
BEACON (WS)	0.533	0.688	0.702	0.996	0.023
Llama	0.430	0.659	0.595	0.991	0.032
Jellyfish	0.423	0.692	0.691	0.992	0.016
RoDo-EM	**0.714**	**0.854**	**0.810**	**0.997**	**0.008**

(b) WDC Product Corpus

Model	Min. ↑	Macro ↑	Weighted ↑	F1 Entropy ↑	PPVP Var. ↓
GEN (Ditto)	0.814	0.864	**0.874**	0.999	0.001
SPEC (Ditto)	0.780	0.848	0.847	0.999	0.005
FlexER	0.774	0.834	0.834	0.999	0.002
BEACON (WS)	0.812	0.868	0.868	0.999	0.002
Llama	0.711	0.766	0.767	0.999	0.001
Jellyfish	0.789	0.836	0.837	0.999	0.003
RoDo-EM	**0.859**	**0.869**	**0.874**	**1.000**	**0.000**

(c) Abt-Buy

Model	Min. ↑	Macro ↑	Weighted ↑	F1 Entropy ↑	PPVP Var. ↓
GEN (Ditto)	0.333	0.796	**0.810**	0.993	0.043
SPEC (Ditto)	0.000	0.163	0.252	0.808	0.028
FlexER	0.000	0.626	0.688	0.941	0.096
BEACON (WS)	0.364	0.732	0.772	0.988	0.061
Llama	0.167	0.388	0.366	0.967	0.020
Jellyfish	0.500	0.793	0.797	0.991	0.011
RoDo-EM	**0.667**	**0.884**	**0.810**	**0.996**	**0.003**

model, hyperparameters, and training data as RoDo-EM to ensure a fair comparison.

GEN (General EM). GEN is the standard general-purpose EM model obtained by fine-tuning a PLM on the union of the training data from all domains. It serves as the primary baseline for evaluating whether robust optimization improves the performance of a single model trained across multiple domains.

SPEC (Domain-Specific EM). SPEC trains one PLM for each domain using only the samples belonging to that domain. While this allows each model to specialize to its corresponding domain, it requires maintaining multiple models and may underperform when individual domains contain limited labeled data.

FlexER [21]. FlexER is a recent PLM-based EM framework that combines transformer representations with a Graph Neural Network (GNN) to support multiple matching intents. For our evaluation, we train FlexER using two intents: (1) the standard EM labels and (2) a coarse-grained product-type intent that is distinct from the domain partitions used in REMAD. This provides a strong graph-based baseline for evaluating domain-aware EM.

BEACON [44, 45]. BEACON is a recent framework for EM across domains that proposes distribution-aware sampling strategies to improve domain-level performance under a fixed labeling budget. While the original BEACON framework trains domain-specific EM models, a recently introduced general-purpose (i.e., domain-agnostic) variant [45] adapts these sampling strategies to train a single EM model across all domains. We use this general-purpose variant as the primary baseline for comparison with RoDo-EM, employing the default 70% downsampling rate from the original work.

We additionally compare RoDo-EM against several prompt-based EM methods, including those built upon LLMs. Following Peeters *et al.* [42], we adopt the *general-complex* prompting strategy and extend it with prompts tailored to each robustness objective (e.g., emphasizing the worst-performing domain or the smallest domain). We evaluate two representative methods:

Llama 3.1 [15], an open-source general-purpose LLM with approximately 8B parameters.

Jellyfish [67], an LLM fine-tuned for data management tasks, including EM.

6 RESULTS AND ANALYSIS

In this section, we evaluate the RoDo-EM framework and analyze its performance on the REMAD problem setting across multiple domain-partitioned datasets and robustness objectives. Section 6.1 presents the primary experimental results. Section 6.2 compares RoDo-EM with prompt-based approaches for improving robustness. Section 6.3 investigates the effect of varying k in the $\mathcal{L}_{\text{worst-}k}$ optimization objective. Finally, Section 6.4 presents a real-world case study demonstrating the practical benefits of RoDo-EM.

6.1 Main Results

We begin by evaluating RoDo-EM across the five robustness metrics introduced in Table 1 and comparing its performance against

Table 3: Ablation of RoDo-EM optimization objectives on WDC-MD.

Method	Target Objective	Target Metric		Relative Improvement (%) Across Metrics							% Domains Improved
		Base	Robust	Worst F1	Macro F1	Weighted F1	F1 Entropy	F1 Var. Reduction	PPVP Var. Reduction	TPRP Var. Reduction	
Worst-1	Minimum Domain F1	0.421	**0.706**	*+67.7*	+3.6	-0.6	+0.5	+56.5	+26.9	+2.1	58.3% (7/12)
DRO	Minimum Domain F1	0.421	**0.714**	*+69.7%*	+0.3	-3.5	+0.5	+63.7	+13.8	+22.6	50% (6/12)
Uniform	Macro F1	0.815	**0.854**	+58.3	*+4.8%*	+0.0	+0.4	+52.2	+59.2	+45.2	75% (9/12)
Entropy	F1 entropy	0.993	**0.997**	+25.0	+0.4	-5.0	*+0.4*	+52.2	+43.4	+50.8	41.7% (5/12)
F1 Variance	F1 variance	0.022	**0.008**	+58.8	+2.5	-3.3	+0.5	*+63.6%*	+55.0	+59.2	58.3% (7/12)
PPVP	PPVP variance	0.026	**0.008**	+40.6	-1.5	-6.0	+0.3	+44.4	*+69.2%*	+11.3	50.0% (6/12)
TPRP	TPRP variance	0.022	**0.009**	+42.1	-1.2	-8.3	+0.3	+35.4	+13.4	*59.1%*	50.0% (6/12)

several modern EM baselines. The reported metrics include minimum domain F1, macro F1, weighted F1, F1 entropy, and PPVP variance. For each metric, we configure RoDo-EM to optimize the corresponding objective by selecting the appropriate loss function (e.g., $\mathcal{L}_{\text{DRO}}$ for min-F1, $\mathcal{L}_{\text{uniform}}$ for macro F1, etc.), followed by a small grid search over the associated robustness weight, sampling strategy, and regularization hyperparameters. We then evaluate how effectively each optimization objective improves its target metric while influencing the remaining evaluation metrics.

Raw Performance. Table 2 summarizes the raw performance of all methods on WDC-MD, the WDC Product Corpus, and Abt-Buy. Across all three datasets, RoDo-EM achieves the strongest performance across each of the evaluation metrics. These improvements stem from jointly optimizing robustness-aware loss functions, employing batch sampling techniques to improve domain exposure during training, and incorporating the regularization techniques described in Section 4.3. Although the magnitude of improvement varies across datasets, consistent gains are observed for the primary robustness objectives. It should be noted that the GEN (Ditto) baseline and the weighted-F1 variant of RoDo-EM achieve identical weighted F1 scores for each dataset because optimizing weighted F1 corresponds to the standard training objective, $\mathcal{L}(\theta)$, used by the underlying DITTO model.

On WDC-MD (Table 2a), RoDo-EM achieves a substantial improvement in minimum domain F1, outperforming the next-best baseline by 34.0%. Improvements in the remaining robustness metrics are more modest but still significant, including a 4.8% increase in macro F1. Domain parity also improves, with PPVP variance reduced from 0.016 to 0.008 compared to the next-best baseline, accompanied by a smaller improvement in F1 entropy. These results suggest that RoDo-EM is particularly effective when training data exhibit substantial domain imbalance.

Table 2b shows more modest improvements across all evaluation metrics. Optimizing minimum domain F1 yields a 5.5% improvement over the next-best baseline. Because the WDC Product Corpus contains a more uniformly distributed set of training data across domains, both F1 entropy and PPVP variance exhibit only marginal improvements, as parity is already high prior to optimization. The macro F1 score obtained by $\mathcal{L}_{\text{uniform}}$ is nearly identical to that achieved by the BEACON baseline, suggesting that the advantages of RoDo-EM are more apparent when domain imbalance is present.

Finally, Table 2c demonstrates that the benefits of RoDo-EM are again substantial on Abt-Buy. Minimum domain F1 improves by 33.4% over the next-best baseline, while macro F1 increases by 11.1%. Although F1 entropy improves only slightly compared to the Jellyfish LLM, RoDo-EM noticeably reduces PPVP variance, decreasing it from 0.011 to 0.003. The textual nature of Abt-Buy enables the Jellyfish LLM to outperform many of the remaining baselines; however, incorporating robustness-aware optimization within RoDo-EM produces further improvements in both robustness and domain parity.

Optimization Objectives. Table 3 presents an ablation study of each optimization objective within the RoDo-EM framework. For each objective, we report the improvement in its target metric, the corresponding changes in the remaining robustness metrics (with target improvement % italicized), and the percentage of domains whose performance either improves or remains unchanged. This analysis characterizes the tradeoffs associated with optimizing different notions of robustness. Italicized values denote the relative improvement achieved in the objective's target metric.

The results show that optimizing for the worst-performing domains yields substantially larger gains in the target metric than the accompanying degradation in average-case performance. For example, $\mathcal{L}_{\text{worst-1}}$ and $\mathcal{L}_{\text{DRO}}$ improve the minimum domain F1 score by 67.7% and 69.7%, respectively, while reducing weighted F1 by only 0.6% and 3.5%. Interestingly, both objectives also improve the domain parity metrics as a secondary effect. This behavior is expected, as raising the performance of the weakest domains while maintaining comparable performance on the remaining domains naturally produces a more balanced distribution of performance across domains. Although weighted F1 decreases modestly relative to the substantial gains in minimum domain performance, this tradeoff remains an important consideration for practitioners when selecting an optimization objective.

The parity-based objectives in Table 3 exhibit a similar tradeoff. Optimizing F1 entropy, F1 variance, PPVP variance, or TPRP variance decreases weighted F1 by roughly 3-8% while producing improvements in the corresponding target metric. Furthermore, optimizing PPVP variance and TPRP variance reduces macro F1 by 1.5% and 1.2%, respectively. Intuitively, encouraging greater parity across domains often requires sacrificing some performance on stronger domains in order to improve weaker ones. Consistent with this interpretation, all four parity objectives also increase the minimum domain F1 score by approximately 25-50%, indicating that improving parity generally benefits the lowest-performing domains. Interestingly, the objective explicitly optimizing a given parity metric does not always produce the largest improvement in that metric, although the differences are generally small. For example, $\mathcal{L}_{\text{DRO}}$

slightly outperforms the F1 variance objective on F1 variance, while optimizing F1 variance yields the largest reduction in TPRP variance. This suggests that the robustness objectives–especially those based on parity–are closely related, with optimizing one objective often yielding comparable improvements across related metrics.

Among the evaluated objectives, $\mathcal{L}_{\text{uniform}}$ exhibits the fewest apparent tradeoffs. Although its target improvement is more modest, increasing macro F1 by 4.8%, it does not degrade any of the remaining evaluation metrics. Moreover, it achieves the highest percentage of improved domains, with 75% (9 of 12) exhibiting improved performance. This behavior closely aligns with the design of $\mathcal{L}_{\text{uniform}}$, which assigns equal importance to every domain during optimization. Overall, these results demonstrate that different robustness objectives provide practitioners with distinct and interpretable tradeoffs, allowing RoDo-EM to be tailored to the requirements of a particular application.

6.2 Prompt-Based Robustness Optimization

We next examine prompt-based robustness as an alternative strategy for REMAD. Rather than directly optimizing a robustness objective during model training, as in RoDo-EM, prompt-based methods can incorporate the desired objective into the natural-language instructions provided to an LLM. This experiment evaluates whether explicitly prompting an EM model to prioritize a particular notion of robustness produces improvements comparable to those achieved through training-time optimization.

We evaluate two open-source LLMs: Llama [15] and Jellyfish [67]. As described in Section 5.3, Llama is a general-purpose LLM, whereas Jellyfish is fine-tuned for data integration tasks, including EM. For both models, we use the *general-complex* zero-shot prompting pattern as the base prompt and introduce small objective-specific modifications for the robust variants. For example, the base prompt for minimum-domain robustness includes two entity descriptions followed by the question:

> "Do the two entity descriptions refer to the same real-world entity?"

A corresponding robust prompt appends the instruction:

> "Optimize especially for the domains that are expected to be hardest or lowest-performing."

This formulation tests whether the model can use knowledge acquired during pretraining or task-specific fine-tuning to infer which candidate pairs belong to inherently difficult or ambiguous domains.

We evaluate prompt-based robustness with respect to minimum domain F1, macro F1, and F1 variance on the same three datasets used in the main experiments. Table 4 reports the relative change in each metric between the base and robustness-aware variants of Llama and Jellyfish, together with the corresponding improvement achieved by RoDo-EM. Positive values indicate improvements, while negative values indicate degradation. For variance reduction, larger positive values indicate a greater decrease in F1 variance.

On WDC-MD, prompt-based robustness produces limited and inconsistent gains. Llama improves minimum domain F1 by only 0.2%, while its macro F1 decreases by 1.5% and its F1 variance increases by 17.4%. Jellyfish improves macro F1 by 2.2% and reduces variance by 68.0%, but its minimum domain F1 decreases by 8.7%. In contrast, RoDo-EM improves each metric, increasing minimum domain F1 by 69.7% and macro F1 by 4.8%, while reducing F1 variance by 63.6%.

A similar pattern appears on the WDC Product Corpus. Robust prompting substantially reduces variance for Llama by 40.5%, but decreases its minimum domain and macro F1 scores by 17.7% and 11.7%, respectively. Jellyfish improves macro F1 by 3.1%, but decreases minimum domain F1 by 5.4% and increases F1 variance by 104.0%. RoDo-EM is again the only method that improves all three metrics, although its gains in minimum domain and macro F1 are comparatively modest because the base model already exhibits relatively uniform domain-level performance on this dataset.

Prompt-based robustness is most effective on Abt-Buy. Llama improves all three metrics, increasing minimum domain F1 by 44.0% and macro F1 by 38.5%, while reducing F1 variance by 43.8%. RoDo-EM achieves the largest improvement in minimum domain F1, increasing it by 100.0%, while also improving macro F1 by 11.1% and reducing variance by 31.9%. Jellyfish exhibits smaller and less consistent changes, leaving minimum domain F1 unchanged, decreasing macro F1 by 0.6%, and reducing variance by 17.1%. The stronger prompt-based results on Abt-Buy may be attributable to its comparatively textual schema, which may provide LLMs with clearer semantic cues for identifying difficult or ambiguous matches.

Overall, our results indicate that the proposed robustness-aware prompting strategy can improve domain-level performance in some settings, although the observed gains are inconsistent and appear to be more effective on text-rich datasets. In contrast, RoDo-EM consistently improves all three robustness measures across the evaluated benchmarks, demonstrating that directly incorporating robustness objectives into model training provides a more reliable mechanism for REMAD than the prompting strategy considered in this study. Nevertheless, these findings are specific to the proposed prompting strategy, and the improvements observed suggest that alternative strategies may provide a more effective means of enforcing robustness in LLM-based EM, making this a promising direction for future work.

6.3 Top-k Robustness Optimization

In Section 4.1, we define the worst-domain objective $\mathcal{L}_{\text{worst-}k}$ for arbitrary values of k. This flexibility is useful when a practitioner wishes to prioritize several challenging or underrepresented domains rather than only the single worst-performing domain. We therefore evaluate how varying k affects both lower-bound and average-case performance.

We conduct this experiment on WDC-MD because it contains the largest number of domains among our benchmarks, with $|\mathcal{D}| = 12$. We evaluate seven values of k in the range $[1, 7]$. For each value, we perform a small hyperparameter search over the robustness weight γ, EMA smoothing, sampler choice, and L2 weight decay. In particular, γ controls the additional weight assigned to the losses of the current worst-k domains.

Figure 5 reports the average F1 score of the k lowest-performing domains, denoted Min-k F1, together with macro and weighted F1. As k increases, Min-k F1 also increases because the metric is averaged over a progressively larger set that includes less challenging domains. However, the gains begin to plateau for approximately

Table 4: Relative improvement (%) of RoDo-EM and prompt-based robustness optimization over their respective base models across each benchmark.

(a) WDC-MD (50% cc, 50% unseen)

Model	Min. Δ	Macro Δ	Var. Red.
RoDo-EM	**+69.7%**	**+4.8%**	**+63.6%**
Llama	+0.2%	−1.5%	−17.4%
Jellyfish	−8.7%	+2.2%	−68.0%

(b) WDC Product Corpus

Model	Min. Δ	Macro Δ	Var. Red.
RoDo-EM	**+5.5%**	+0.6%	**+94.9%**
Llama	−17.7%	−11.7%	+40.5%
Jellyfish	−5.4%	**+3.1%**	−104.0%

(c) Abt-Buy

Model	Min. Δ	Macro Δ	Var. Red.
RoDo-EM	**+100.0%**	+11.1%	+31.9%
Llama	+44.0%	**+38.5%**	**+43.8%**
Jellyfish	+0.0%	−0.6%	+17.1%

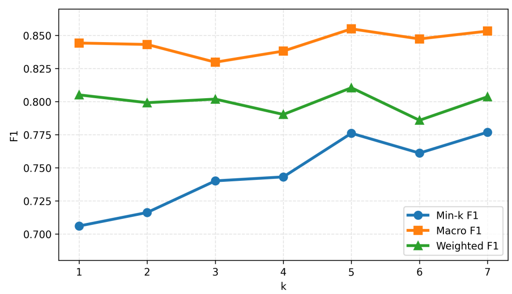

Figure 5: Minimum-k, macro, and weighted F1 scores across robust Top-k RoDo-EM models.

k = 5-7, suggesting diminishing benefits from emphasizing an increasingly large subset of domains. As k approaches $|\mathcal{D}|$, $\mathcal{L}_{\text{worst-}k}$ increasingly resembles uniform domain weighting and therefore loses its ability to target the weakest domains.

Macro and weighted F1 remain comparatively stable across values of k, with macro F1 ranging from $[0.830, 0.855]$ and weighted F1 ranging from $[0.786, 0.811]$. This stability indicates that practitioners can vary k to control how broadly robustness is distributed across weaker domains without substantially changing average-case performance. Overall, smaller values of k provide more targeted optimization, whereas larger values distribute the robustness emphasis across a broader set of domains.

6.4 Case Study

We conclude our evaluation with a practical case study based on the Walmart–Amazon EM benchmark. This case study builds on the motivating example from Section 1, in which a practitioner at a large e-commerce firm must process a substantial EM workload of product pairs partitioned into price-based domains. Although lower-priced products account for a larger proportion of the workload, errors involving higher-priced products may carry greater practical or financial consequences.

We partition the Walmart-Amazon candidate pairs into the same three price ranges introduced in Example 1.1: *low*, *medium*, and *high*. The benchmark's training set contains 6,144 product pairs, with product prices ranging from <$10 to >$20,000. We assign each pair with a price equal to the average of the two product prices. When a price is available for only one product in the pair, we use that value instead. We define low-priced pairs as those with a price below $50, medium-priced pairs as those priced from $50 to $300, and high-priced pairs as those priced above $300. These thresholds produce an approximately 2:1:1 distribution across the three price domains. We apply the same partitioning procedure to the validation and test sets, which exhibit similar domain proportions.

Robustness is particularly important in this setting because the medium and high-priced domains each contain fewer training examples than the low-priced domain, yet their matching decisions may have greater monetary significance. We therefore consider a practitioner whose deployment requirement is to achieve an F1 score of at least 0.75 in every price domain. To evaluate whether this requirement is satisfied, we define *threshold domains* as the proportion of test domains whose F1 score meets or exceeds 0.75. Unlike aggregate F1 alone, this metric allows the practitioner to verify that the model performs adequately on the smaller medium and high-priced domains rather than obtaining strong overall performance primarily from the dominant low-priced domain.

To meet this requirement, the practitioner trains RoDo-EM using the $\mathcal{L}_{\text{DRO}}$ objective, which prioritizes poorly performing domains and seeks to maximize the minimum domain-level F1 score. Table 5 reports the resulting F1 score for each price domain, weighted F1, the percentage of threshold domains, and the *recovered value*. We define recovered value as the sum of the representative prices of all true-match pairs that the model correctly predicts as matches. This metric provides a simple proxy for the monetary value associated with correctly identifying matching products, such as enabling the corresponding products to be listed, advertised, or otherwise processed by the e-commerce platform.

RoDo-EM exhibits a decrease in low-price F1, from 0.8198 to 0.8052, but improves performance in both of the more valuable price domains. Medium-price F1 increases from 0.7970 to 0.8000, while high-price F1 increases substantially from 0.7097 to 0.7857. Consequently, weighted F1 also improves from 0.7892 to 0.7994 despite the modest reduction in the low-price domain. More importantly, RoDo-EM satisfies the practitioner's minimum-performance requirement in all three domains, yielding 100% threshold-domain coverage, compared with 66.7% for the base DITTO model, which fails to reach the required F1 score for high-priced products.

These domain-level improvements also translate into greater recovered value. RoDo-EM recovers $22,885.68, representing an increase of $411.25, or 1.83%, over the base model. Although this case study represents a simplified deployment scenario, it illustrates how optimizing lower-bound domain performance can produce benefits that are not fully captured by aggregate evaluation metrics. In particular, RoDo-EM allows practitioners to impose application-specific performance requirements across operationally important domains while maintaining strong overall matching performance. These results also demonstrate the practical value of REMAD in

Table 5: Price-Aware Entity Matching Case Study

Model	High Price F1	Medium Price F1	Low Price F1	Weighted F1	Threshold Domains	Recovered Value
Base Model	0.7097	0.7970	**0.8198**	0.7892	66.7% (2/3)	$22,474.43
RoDo-EM	**0.7857**	**0.8000**	0.8052	**0.7994**	**100% (3/3)**	**$22,885.68**
Δ	+0.0760	+0.0030	−0.0146	+0.0103	+33.3%	+$411.25 (+1.83%)

settings where particular domains may be critical to downstream business objectives.

7 CONCLUSION

This paper introduced *Robust Entity Matching Across Domains* (REMAD), a new problem setting that extends traditional domain-aware Entity Matching by explicitly optimizing domain-level robustness. To address this problem, we presented **RoDo-EM**, a modular optimization framework that integrates robustness-aware objectives, domain-aware sampling strategies, and regularization techniques into the training pipeline of EM models. Through extensive experiments on multiple domain-partitioned benchmarks, we demonstrated that RoDo-EM consistently improves lower-bound domain performance and domain parity while maintaining competitive overall matching performance, outperforming existing PLM-based and prompt-based EM approaches across a variety of robustness objectives. Furthermore, our analyses showed that the framework integrates a suite of distinct objectives with configurable and interpretable tradeoffs between robustness, parity, and average-case performance. In future work, RoDo-EM could be extended to larger backbone models, including fine-tuned LLMs and emerging foundation models for data management. Further, future work could investigate adaptive optimization techniques that automatically adjust robustness weights and sampling strategies throughout training. More broadly, we hope this work motivates further research on robust optimization techniques for domain-aware data integration tasks beyond Entity Matching.

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Table 6: Sensitivity of RoDo-EM to the robustness weight γ on WDC-MD.

Gamma Value	$\mathcal{L}_{\text{DRO}}$		$\mathcal{L}_{\text{worst-1}}$		$\mathcal{L}_{\text{par}}(m_{\text{PPVP}})$	
	Min. F1	Weighted F1.	Min. F1	Weighted F1	PPVP Var.	Weighted F1
$\gamma = 1.0$	0.6250	0.7642	0.6316	0.7952	0.0199	**0.7861**
$\gamma = 1.5$	0.6667	0.7515	0.7000	0.7850	0.0196	0.7663
$\gamma = 2.0$	0.6667	0.7643	0.6957	0.7829	0.0172	0.7702
$\gamma = 2.5$	0.6667	0.7663	0.6531	0.7747	0.0126	0.7780
$\gamma = 3.0$	0.5833	0.6964	0.5882	0.7312	0.0180	0.7362
$\gamma = 3.5$	0.5882	0.7521	0.6667	0.7648	0.0131	0.7788
$\gamma = 4.0$	**0.7143**	**0.7774**	**0.7059**	**0.8008**	**0.0080**	0.7627
$\gamma = 4.5$	0.6316	0.7579	0.5263	0.7848	0.0141	0.7341
$\gamma = 5.0$	0.6596	0.7550	0.6154	0.7720	0.0135	0.7181

Table 7: Impact of sampling methods on $\mathcal{L}_{\text{worst-1}}$ optimization on WDC-MD.

Sampling Method	Min. F1	Macro F1	Weighted F1
None	0.6154	0.7927	0.7355
Balanced	0.5882	0.8129	0.7765
Loss-based	**0.7059**	**0.8446**	**0.8008**

A Hyperparameter Sensitivity

In this section, we examine the two primary hyperparameters of the RoDo-EM framework: the robustness weight γ (Section 4.1) and the domain sampling strategy (Section 4.2). Both components directly influence the optimization process, and our goal is to understand how they affect the tradeoff between the target robustness objective and overall EM performance.

To study the effect of γ, we evaluate three representative optimization objectives: $\mathcal{L}_{\text{DRO}}$, $\mathcal{L}_{\text{worst-1}}$, and $\mathcal{L}_{\text{PPVP}}$. For each objective, all hyperparameters are fixed to the values obtained during the grid search used for the main experiments, while only the corresponding robustness weight (γ_{DRO}, $\gamma_{\text{worst-1}}$, or γ_{par}) is varied. We evaluate γ values from 1.0 to 5.0 in increments of 0.5 and report the corresponding target metric (minimum F1 for DRO and Worst-1, PPVP variance for $\mathcal{L}_{\text{PPVP}}$) together with weighted F1 in Table 6.

The results indicate that moderate robustness weights generally provide the best tradeoff, although the relationship between γ and performance is not monotonic. For both $\mathcal{L}_{\text{DRO}}$ and $\mathcal{L}_{\text{worst-1}}$, $\gamma = 4.0$ achieves the strongest combination of minimum-domain and weighted F1 performance. However, nearby values such as $\gamma = 2.5$ for $\mathcal{L}_{\text{DRO}}$ and $\gamma = 1.5$ for $\mathcal{L}_{\text{worst-1}}$ achieve comparable results, suggesting that the framework is relatively robust to moderate changes in γ. In contrast, $\mathcal{L}_{\text{PPVP}}$ exhibits a clearer tradeoff: larger values of γ improve parity, with the lowest PPVP variance achieved at $\gamma = 4.0$, while smaller values such as $\gamma = 1.0$ produce the highest weighted F1. These results suggest that practitioners should select γ based on whether domain parity or overall predictive performance is the primary deployment objective.

We next investigate the effect of the domain-aware batch sampler. Table 7 compares three sampling strategies for $\mathcal{L}_{\text{worst-1}}$: the default random sampler, a balanced sampler that enforces approximately equal domain representation within each batch, and a loss-based sampler that oversamples domains with higher losses from previous training iterations. All remaining hyperparameters are fixed to the values used in the main experiments.

The loss-based sampler consistently produces the strongest results, improving not only minimum domain F1 but also macro and weighted F1 relative to the default sampler. This suggests that emphasizing difficult domains during training encourages the model to learn more informative representations that generalize beyond those domains. The balanced sampler also improves macro and weighted F1, but slightly decreases minimum domain F1, indicating that simply equalizing domain frequencies is less effective than explicitly prioritizing challenging domains. Overall, these results demonstrate that domain-aware sampling is an important component of RoDo-EM, with loss-based sampling providing the most effective balance between robustness and overall EM performance.